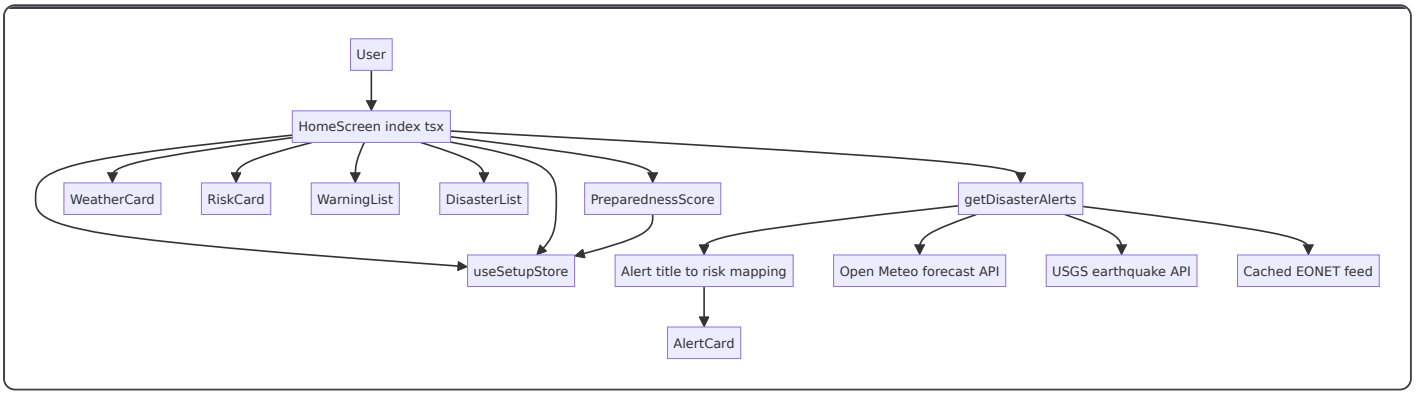
Emergency and Inventory Management Domain

Overview

This dashboard section is the app's landing experience for disaster awareness. It combines live alert synthesis, weather conditions, regional location context, survival guidance, and a preparedness score into one scrollable home screen.

The screen logic in `app/(tabs)/index.tsx` is intentionally thin: it reads persisted setup state from `useSetupStore`, fetches live disaster data through `getDisasterAlerts`, converts returned alert titles into card-ready risk values, and then hands display concerns to reusable widgets such as `WeatherCard`, `RiskCard`, `AlertCard`, `DisasterList`, `WarningList`, and `PreparednessScore`. The result is a clear composition boundary: the screen coordinates data, while the components render presentation.

Architecture Overview



Home Dashboard Orchestration

Home Screen

> **IMPORTANT**
> The home dashboard is driven by `coords` and `location` from `useSetupStore`. Until coordinates exist and `getDisasterAlerts` resolves, the screen returns `null` and renders nothing.

`disaster-app/app/(tabs)/index.tsx`

The `HomeScreen` function is the composition root for this section. It reads `coords` and `location` from `useSetupStore`, triggers live data loading in a `useEffect`, and renders the dashboard only after data has arrived.

Responsibilities

- Reads persisted location context from `useSetupStore`.
- Calls `getDisasterAlerts(coords.lat, coords.lon)` when coordinates are available.
- Derives alert card entries from `data.alerts` and `data.risk`.
- Renders the dashboard widgets in a fixed vertical order.
- Surfaces the current district and city above the weather card.

Local State and Derived Values

Name	Type	Purpose
`data`	`any`	Holds the resolved object returned by `getDisasterAlerts`.
`alertsForCard`	`{ title: string; riskLevel: number }[]`	Derived from `data.alerts` and mapped to component-specific risk values.
`TEST_DISASTER`	`null`	Test override slot; currently inactive.

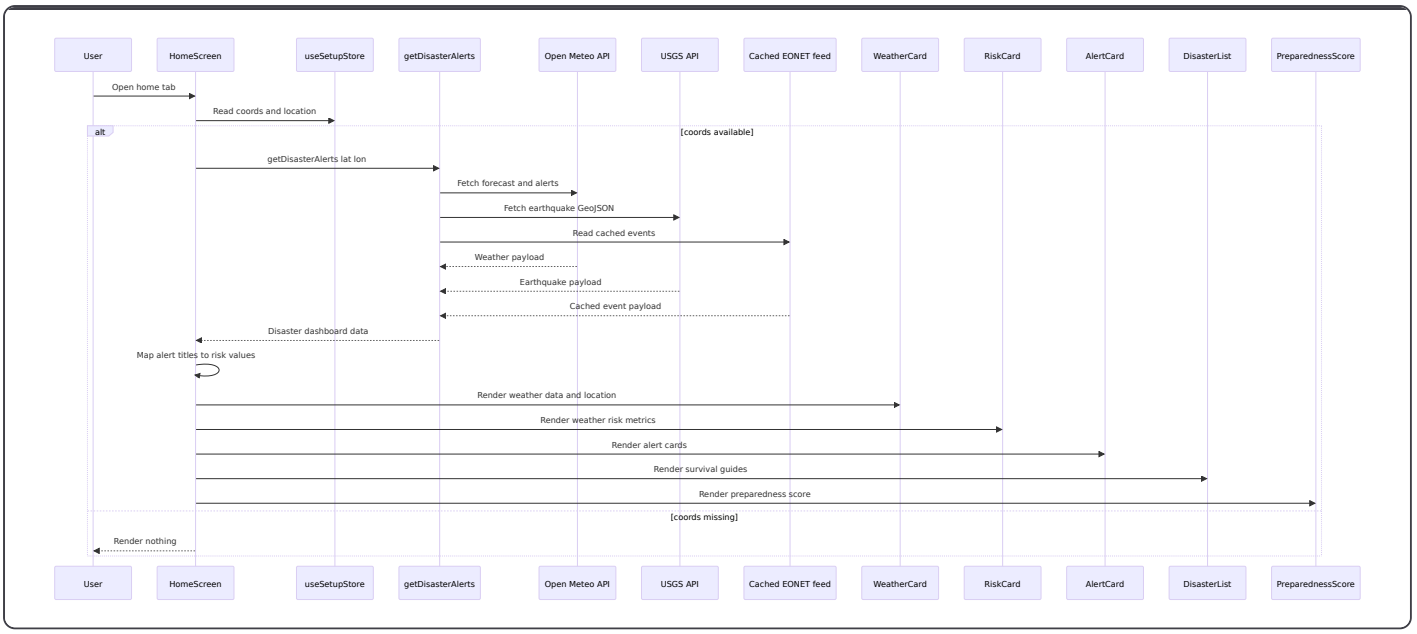
Data-to-UI Mapping

Incoming alert title	Risk source
`Flood Risk`	`data.risk.flood`
`Storm Risk`	`data.risk.storm`
`Heatwave Risk`	`data.risk.heat`
`Earthquake Risk`	`data.risk.earthquake`
Any other title	`0`

Render Order

1. Location header using `location?.district` and `location?.city`
2. `WeatherCard` with `data.weather` and `location`
3. `RiskCard` with full `data`
4. `AlertCard` with `alertsForCard`
5. `DisasterList`
6. `PreparednessScore`

Home Screen Data Fetch Flow



Presentation Widgets

Weather Card

> **NOTE**
> `TEST_DISASTER` is hard-coded to `null`, so the override branch that assigns fixed risk values never runs. [!NOTE] `getDisasterAlerts` already catches its own failures and returns a fallback payload, so the `HomeScreen` `.catch(console.error)` path only applies if the helper rejects unexpectedly.

`disaster-app/components/WeatherCard.tsx`

`WeatherCard` renders the live weather summary that the home screen receives from `getDisasterAlerts`. It is driven by localized labels from `useLanguageStore` and uses the location district to anchor the current reading to a place.

Props

Property	Type	Description
`weather`	`any`	Weather object containing `temp`, `wind`, `humidity`, and `weatherType`.
`location`	`any`	Location object used to display the district.

Public Methods

Method	Description
`getWeatherIcon`	Maps `weatherType` strings to `Ionicons` glyph names.

Weather Icon Mapping

Weather type input	Icon
Contains `clear` or equals `sunny`	`sunny`
Contains `partly cloudy`	`cloudy`
Contains `cloud`	`cloud`
Contains `fog`	`cloudy-night`
Contains `drizzle`	`rainy`
Contains `rain`	`rainy`
Contains `snow`	`snow`
Contains `freezing`	`ice-cream`
Contains `thunderstorm`	`thunderstorm`
Contains `wind`	`wind`
Contains `hot`	`thermometer`
Anything else	`partly-sunny`

Rendered Content

- Current temperature in degrees Celsius.
- Wind and humidity line with localized labels.
- Location district.
- Weather icon derived from `weather.weatherType`.

Risk Card

`disaster-app/components/Riskcard.tsx`

`RiskCard` is a compact weather metrics strip. Despite the filename, the component does not render disaster risk percentages; it renders weather conditions from `data.weather`.

Props

Property	Type	Description
`data`	`any`	Full dashboard object returned by `getDisasterAlerts`.
`data.weather`	object	Weather payload containing humidity, wind, and rain.

Public Methods

Method	Description
`RiskCard`	Renders the weather metric row from `data.weather`.

Rendered Metrics

Metric	Source field	Display format
Humidity	`weather.humidity`	Percentage
Wind	`weather.wind`	`km/h`
Rain	`weather.rain`	`mm`, or `–` when undefined

Alert Card

`disaster-app/components/AlertCard.tsx`

`AlertCard` turns normalized alert entries into visual severity cards. It localizes the empty state and severity badges through `useLanguageStore`.

Props

Property	Type	Description
`alerts`	`AlertItem[]`	Array of alert entries with title and computed risk level.

Public Methods

Method	Description
`getRiskBackground`	Chooses a background color based on `riskLevel`.
`getAlertIcon`	Maps an alert title to an icon name and icon color.
`AlertCard`	Renders alert cards or the empty state.

`AlertItem` Interface

Property	Type	Description
`title`	`string`	Alert title such as `Flood Risk`.
`riskLevel`	`number`	Normalized risk percentage passed to the card.

Risk Styling Rules

Risk level	Background color	Badge label
`>= 80`	`#FFEBEE`	`HIGH`
`>= 60`	`#FFF3E0`	`HIGH`
`>= 40`	`#FFFDE7`	`MEDIUM`
`< 40`	`#E8F5E9`	`LOW`

Alert Icon Mapping

Title	Icon	Color
`Flood Risk`	`water`	`#1E88E5`
`Storm Risk`	`thunderstorm`	`#5E35B1`
`Heatwave Risk`	`sunny`	`#F39C12`
`Earthquake Risk`	`pulse`	`#E53935`
Anything else	`alert-circle`	`#757575`

Empty State

When `alerts` is empty or missing, the card shows:

- a green `checkmark-circle` icon
- the localized `noActiveAlerts` message

Warning List

`disaster-app/components/WarningList.tsx`

`WarningList` is a reusable early-warning strip built directly from a `risk` object. It presents the risk values as horizontal cards and labels them as either high risk or low risk.

Props

Property	Type	Description
`risk`	`any`	Risk object with `flood`, `storm`, `heat`, and `earthquake` fields.

Public Methods

Method	Description
`WarningList`	Renders the horizontal early warning cards.

Risk Cards

Title	Value source
Flood	`risk.flood`
Storm	`risk.storm`
Heat	`risk.heat`
Earthquake	`risk.earthquake`

Severity Label Rule

Value	Label
`> 60`	`High Risk`
`<= 60`	`Low`

Disaster List

`disaster-app/components/DisasterList.tsx`

`DisasterList` exposes survival guides as tappable icons. It sorts disaster types by regional score, opens a modal for the selected disaster, and renders localized survival steps.

Public Methods

Method	Description
`getIconName`	Maps a disaster type to a `FontAwesome6` icon name.
`getDisasterScore`	Computes `freq * sev` for a disaster and region.
`DisasterList`	Renders the guide launcher row and the survival modal.

Disaster Types in `SURVIVAL_GUIDE`

Type	stepsKey
Flood	`flood`
Earthquake	`earthquake`
Storm	`storm`
Heat Wave	`heatWave`
Landslide	`landslide`

Icon Mapping

Disaster type	Icon
Flood	`water`
Earthquake	`earth-asia`
Storm	`cloud-rain`
Heat Wave	`sun`
Landslide	`mountain`
Anything else	`triangle-exclamation`

Interaction Model

- The list is horizontal and uses `FlatList`.
- Tapping an icon stores the selected disaster in local state.
- The modal reads localized steps from `stepsData` using the active language and the disaster's `stepsKey`.
- The modal closes via the header close icon, the footer button, or `onRequestClose`.

Preparedness Score

`disaster-app/components/PreparednessScore.tsx`

`PreparednessScore` turns location-aware inventory readiness into a visual percentage ring. It computes the top regional disaster, finds the essential supplies for that disaster, and compares them against stocked inventory items from `useSetupStore`.

Public Methods

Method	Description
`getTopDisaster`	Returns the highest scoring disaster for a region.
`getEssentialItemsForDisaster`	Filters supplies where disaster priority is above `0.7`.
`PreparednessScore`	Renders the score ring, text summary, and stocked essentials count.

Data Sources

Source	Purpose
`Disaster` from `DB/stateDisasterData.json`	Used to determine the top disaster for the current region.
`supplies` from `DB/supplies.json`	Used to find essential items for the selected disaster.
`useSetupStore`	Provides `location` and `items`.
`useLanguageStore`	Localizes labels and count text.

Score Calculation

Step	Expression	
Determine top disaster	Highest `(freq * sev)` for `location.region`	
Select essentials	`supplies.filter((item) => (item.priority?.[disasterType]	0) > 0.7)`
Count stocked essentials	Compare stocked item names from `items` against `item.item` values in supplies	
Compute score	`(stockedEssentialCount / totalEssential) * 100`	

Color Thresholds

Score	Color
`< 50`	`#EF4444`
`< 75`	`#F59E0B`
`>= 75`	`#10B981`

UI Output

- Circular SVG progress ring.
- Percentage text centered inside the ring.
- Title localized as `Preparedness Score`.
- Subtitle formatted as `for {{disaster}} in {{region}}`.
- Stocked essentials summary such as `3 / 8 essentials stocked`.

Data Integration

Disaster Alerts Utility

`disaster-app/utils/Data.ts`

`getDisasterAlerts` is the dashboard data builder. It fetches weather, earthquake, and cached event data in parallel, converts them into a single summary object, and returns fallback data if parsing or network work fails.

Public Methods

Method	Description
`getDisasterAlerts`	Builds the home dashboard payload for a latitude and longitude.

Runtime Output Shape

Property	Type	Description
`location`	object	Coordinates used for the fetch request.
`location.lat`	`number`	Latitude passed into the data sources.
`location.lon`	`number`	Longitude passed into the data sources.
`weather`	object	Derived weather summary.
`weather.temp`	`number`	Rounded temperature.
`weather.wind`	`number`	Wind speed from the current forecast.
`weather.rain`	`number`	Current precipitation.
`weather.rainProb`	`number`	Current precipitation probability.
`weather.humidity`	`number`	Relative humidity.
`weather.weatherType`	`string`	Normalized weather label.
`alerts`	`string[]`	Alert titles such as `Flood Risk` or official weather event names.
`risk`	object	Computed disaster risk summary.
`risk.flood`	`number`	Flood risk percentage.
`risk.storm`	`number`	Storm risk percentage.
`risk.heat`	`number`	Heat risk percentage.
`risk.earthquake`	`number`	Earthquake risk percentage.

Error Handling Behavior

- Runs weather, earthquake, and cached event reads with `Promise.allSettled`.
- Parses each source inside its own `try/catch`.
- On failure, returns a fallback object with:
  - `temp: 20`
  - `wind: 0`
  - `rain: 0`

- `rainProb: 0`
- `humidity: 50`
- `weatherType: 'Unknown'`
- empty risk scores
- `alerts: ['Unable to fetch disaster data']`

Alert Synthesis Rules

- Flood alerts are only added when `floodRisk >= 70` and the NASA flood source is present.
- Storm alerts are only added when `stormRisk >= 70` and the NASA storm source is present.
- Heat alerts are added when `heatRisk >= 70`.
- Earthquake alerts are added when `earthquakeRisk >= 50`.
- Official weather alerts are appended if they are not already present.

API Integration

Open Meteo Forecast and Alerts

Get Current Forecast and Alerts

{ } **Get Current Forecast and Alerts**

Fetches current weather metrics and official alerts for the dashboard location

GET `https://api.open-meteo.com/v1/forecast`

Query parameters

latitude string required

<lat>

longitude string required

<lon>

current string required

temperature_2m,wind_speed_10m,precipitation,precipitation_probability,relative_humidity_2m,weathercode

alerts string required

true

Code examples

Copy

Curl Javascript Python

```
curl --request GET \
  --url "https://api.open-meteo.com/v1/forecast?latitude=<lat>&longitude=
<lon>&current=temperature_2m,wind_speed_10m,precipitation,precipitation_probability,relative_humidity_2m,
weathercode&alerts=true"
```

Responses

200

Success


```
{
    "current": {
        "temperature_2m": 31.4,
        "wind_speed_10m": 18.2,
        "precipitation": 2.1,
        "precipitation_probability": 64,
        "relative_humidity_2m": 78,
        "weathercode": 61
    },
    "alerts": [
        {
            "event": "Flood Watch"
        }
    ]
}
```


USGS Earthquake GeoJSON Query

Get Recent Earthquakes Near Location

{ } Get Recent Earthquakes Near Location

Fetches nearby earthquake events used to calculate earthquake risk

GET `https://earthquake.usgs.gov/fdsnws/event/1/query`

Query parameters

format string required

geojson

latitude string required

<lat>

longitude string required

<lon>

maxradiuskm string required

500

limit string required

5

Code examples

Copy

Curl Javascript Python

```
curl --request GET \
  --url "https://earthquake.usgs.gov/fdsnws/event/1/query?format=geojson&latitude=<lat>&longitude=
<lon>&maxradiuskm=500&limit=5"
```

Responses

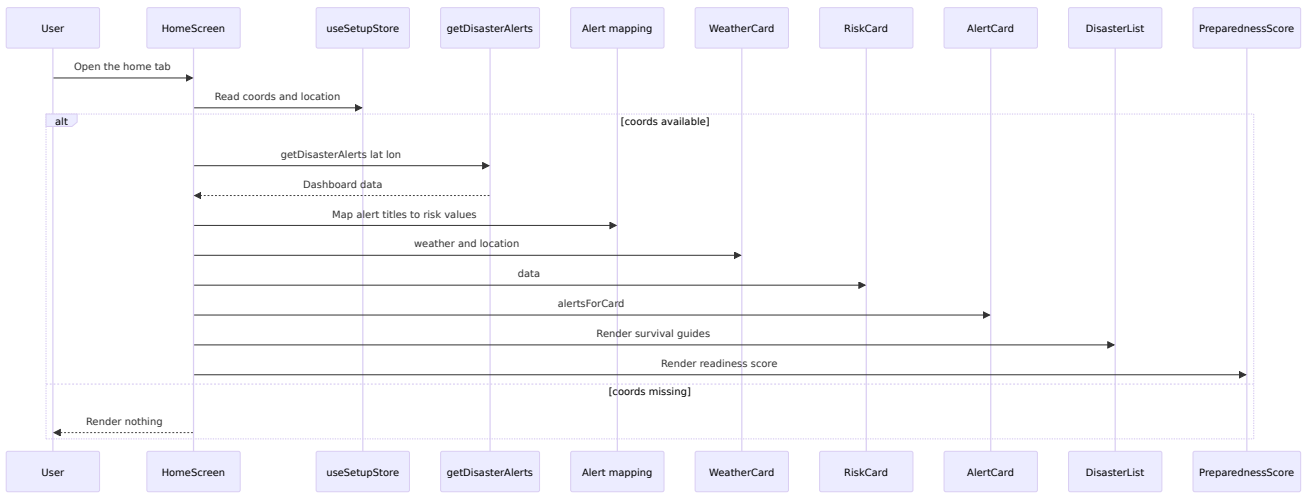
200

Success

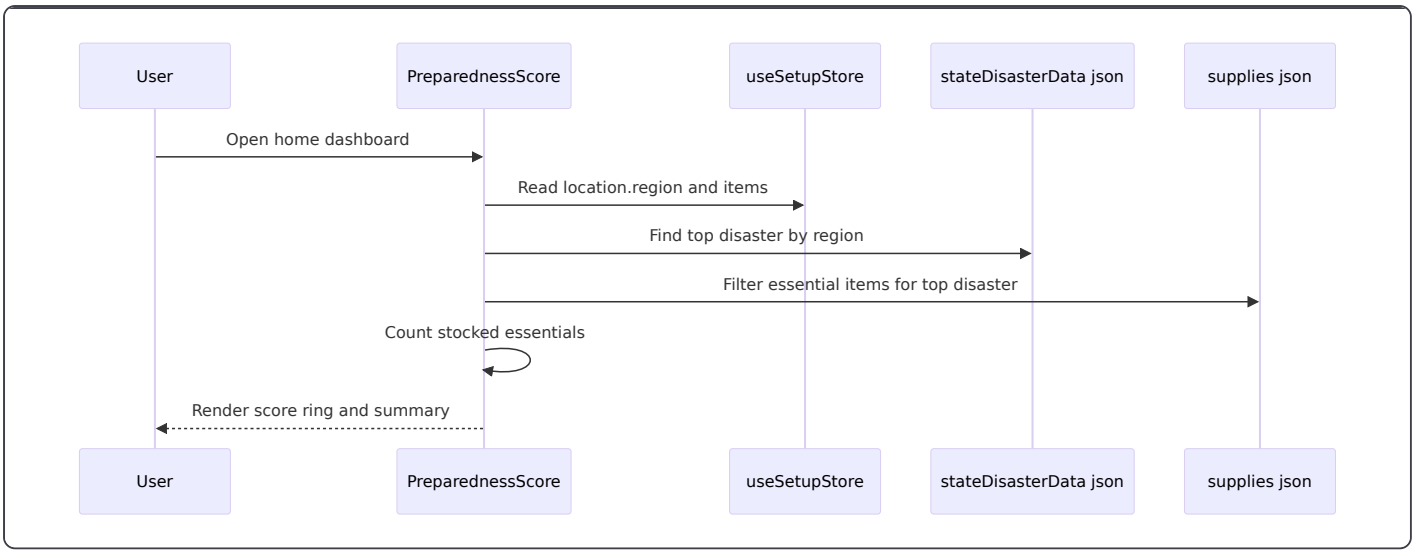
```
{
    "features": [
        {
            "properties": {
                "mag": 5.4
            }
        }
    ]
}
```

Feature Flows

Dashboard Hydration and Alert Rendering



Preparedness Score Recalculation



State Management

Persisted Setup State

The dashboard composes multiple values from `useSetupStore`:

- `coords` drives live data fetching.
- `location` drives the location header, `WeatherCard`, `DisasterList`, and `PreparednessScore`.
- `items` drives preparedness calculations.

Persisted Language State

`useLanguageStore` is used by:

- `WeatherCard`
- `AlertCard`
- `DisasterList`
- `PreparednessScore`

This keeps labels and empty-state text aligned with the active language across the dashboard widgets.

UI States

Component	State	Behavior
`HomeScreen`	No coordinates	Skips fetch and returns `null`.
`HomeScreen`	Data loading	Waits for `getDisasterAlerts` to resolve.
`AlertCard`	Empty alerts	Shows the green checkmark empty state.
`PreparednessScore`	No essentials	Renders a `0%` score.
`DisasterList`	Modal closed	Shows compact disaster icons only.
`DisasterList`	Modal open	Shows localized survival steps for the selected disaster.

Error Handling

`getDisasterAlerts` is the primary fault boundary for the dashboard. Its implementation handles partial failures by settling each upstream request independently, parsing each response behind its own guard, and returning a fallback object instead of throwing.

The dashboard widgets also tolerate missing data:

- `WeatherCard` uses a fallback weather label of `Partly Cloudy` when `weatherType` is missing.
- `AlertCard` renders a no-alert state for empty input.
- `RiskCard` displays `–` for missing rain values.
- `PreparednessScore` returns `0` when there are no essential items for the region.
- `DisasterList` and `PreparednessScore` both default top-disaster logic to `Flood` when no stronger match is found.

Dependencies

Internal Modules

- `useSetupStore` from `utils/setup`
- `useLanguageStore` from `utils/languageStore`
- `getDisasterAlerts` from `utils/Data`
- `stateDisasterData.json`
- `supplies.json`

External Packages

- `react`
- `react-native`

- `expo-status-bar`
- `@expo/vector-icons`
- `react-native-svg`
- `zustand`

API Providers

- Open Meteo
- USGS Earthquake GeoJSON
- Cached EONET event feed through the dashboard utility

Testing Considerations

- Verify that `coords` from `useSetupStore` trigger `getDisasterAlerts` once and only when available.
- Verify that each returned alert title maps to the intended `riskLevel`.
- Verify that `AlertCard` renders the empty state when `alerts` is empty.
- Verify that `PreparednessScore` changes when `items` or `location.region` changes.
- Verify that `DisasterList` opens and closes its modal correctly for each disaster type.
- Verify that localized labels render correctly for both `en` and `ja`.
- Verify that `WeatherCard` uses the location district and weather type icon mapping correctly.
- Verify that the fallback payload from `getDisasterAlerts` still produces a stable dashboard shape.

Key Classes Reference

Class	Responsibility
`index.tsx`	Orchestrates home dashboard data loading and widget composition.
`WeatherCard.tsx`	Renders live weather, temperature, and location context.
`Riskcard.tsx`	Renders humidity, wind, and rain metrics from dashboard weather data.
`AlertCard.tsx`	Renders normalized disaster alert cards with severity badges.
`WarningList.tsx`	Renders a compact horizontal risk summary from the risk object.
`DisasterList.tsx`	Shows survival guides and localized disaster steps in a modal.
`PreparednessScore.tsx`	Calculates and displays preparedness coverage for the top regional disaster.
`Data.ts`	Fetches and synthesizes live disaster dashboard data.